

Supervised Machine Learning Classification

Telecom Customer Churn – Final Report

1.1 Has the learner uploaded the final report as a PDF document that focuses on key insights and findings?

Yes. The final report presents the results of a supervised machine-learning classification analysis for predicting telecom customer churn. The analysis focuses on identifying customers who are likely to leave the telecom service and determining which classification algorithm provides the most suitable predictions. The report summarizes the dataset, preprocessing, classification models, evaluation metrics, model comparison, cross-validation, hyperparameter tuning, and the final model selection. The main finding is that ensemble tree-based methods, particularly Random Forest and Gradient Boosting, performed substantially better than Logistic Regression and AdaBoost on the test data. Among the initial models evaluated, Random Forest achieved approximately **92.8% accuracy and a 0.724 F1-score**, while Gradient Boosting achieved approximately **92.4% accuracy and a 0.712 F1-score**. The analysis also used 5-fold stratified cross-validation. Random Forest achieved a mean cross-validation F1-score of approximately **0.755**, indicating reasonably consistent performance across validation folds. The report focuses not only on accuracy but also on precision, recall, F1-score, ROC-AUC, PR-AUC, confusion matrices, and cross-validation performance. These measures are particularly important because the dataset contains substantially more non-churn customers than churn customers.

1.2 Data Description

The dataset used in this analysis is a **telecom customer churn dataset containing 3,333 customer records and 11 variables**. The target variable is **Churn**, which represents whether a customer left the telecom service. The target contains two classes: **0 = did not churn** and **1 = churned**. There are 2,850 non-churn customers and 483 churn customers, meaning the churn class represents approximately 14.5% of the observations. The dataset is therefore imbalanced. The predictor variables are AccountWeeks, ContractRenewal, DataPlan, DataUsage, CustServCalls, DayMins, DayCalls, MonthlyCharge, AverageFee, and RoamMins. The notebook uses these variables as input features (X), while Churn is used as the target (y). The notebook checks the dataset shape, data types, missing values, duplicates, and class distribution. No missing values were reported, all variables are numeric, and no duplicate rows were detected. The analysis also investigates relationships between features and churn. For example, churned customers had higher average customer-service calls (2.23 versus 1.45), higher daytime minutes (206.91 versus 175.18), and higher monthly charges (59.19 versus 55.82). Churned customers also had a lower contract-renewal rate (0.716 versus 0.935). These patterns indicate that the available customer-behavior variables contain useful information for classification.

1.3 Main Objective

The main objective of this analysis is to develop a **supervised machine-learning classification model capable of predicting whether a telecom customer is likely to churn**. Customer churn is an important business problem because losing customers can reduce revenue and increase the cost of acquiring new customers. Therefore, the purpose of the model is not simply to achieve high overall accuracy, but to identify customers who are at risk of leaving the telecom service. The analysis compares Logistic Regression, Decision Tree, Random Forest, Bagging, AdaBoost, and Gradient Boosting to determine which model provides the best balance between correctly identifying churners and minimizing incorrect predictions. Particular attention is given to precision, recall, F1-score, ROC-AUC, and PR-AUC because the churn class is considerably smaller than the non-churn class. The ultimate goal is to identify customers at risk of churn so that a telecom company could potentially take preventative actions such as targeted retention offers or customer-service interventions.

1.4 Classification Model Comparison

Six supervised classification algorithms were evaluated: **Logistic Regression, Decision Tree, Random**

Forest, Bagging, AdaBoost, and Gradient Boosting. The models were compared using Accuracy, Precision, Recall, F1-score, ROC-AUC, and PR-AUC. The initial test results were:

Logistic Regression: 85.76% accuracy, 0.526 precision, 0.206 recall, 0.296 F1.

Decision Tree: 89.21% accuracy, 0.640 precision, 0.588 recall, 0.613 F1.

Random Forest: 92.80% accuracy, 0.818 precision, 0.649 recall, 0.724 F1.

Bagging: 92.65% accuracy, 0.833 precision, 0.619 recall, 0.710 F1.

AdaBoost: 87.41% accuracy, 0.633 precision, 0.320 recall, 0.425 F1.

Gradient Boosting: 92.35% accuracy, 0.788 precision, 0.649 recall, 0.712 F1. These results show that

Random Forest performed best overall among the initial models, achieving the highest accuracy and F1-score. It also produced strong ROC-AUC and PR-AUC results. Five-fold stratified cross-validation was then performed. Random Forest produced F1-scores of approximately 0.781, 0.738, 0.724, 0.772, and 0.762, with a mean F1-score of approximately **0.755**. This indicates reasonably stable performance across validation folds. Hyperparameter tuning was also performed using GridSearchCV. The best Random Forest parameters were `n_estimators = 100`, `max_depth = None`, `min_samples_leaf = 2`, and `min_samples_split = 2`. The tuned model achieved a cross-validation F1-score of approximately **0.762**. However, its held-out test F1-score was approximately 0.705, slightly below the original Random Forest's 0.724. Therefore, based on the current results, the **original Random Forest is the preferred model for the final analysis**.

1.5 Limitations of the Selected Model

The Random Forest model performs well overall, but several limitations should be considered. **1. Class imbalance:** The dataset contains 2,850 non-churn customers and only 483 churn customers. Because the churn class represents approximately 14.5% of observations, accuracy alone can give an incomplete picture of performance. This is why precision, recall, F1-score, ROC-AUC, and PR-AUC are also considered. **2. Recall is not perfect:** Random Forest achieves approximately 64.95% recall for the churn class. This means that the model identifies many actual churners, but it still misses a meaningful proportion of them. From a business perspective, these false negatives are important because those customers may not receive retention interventions. **3. Limited dataset size:** The analysis uses 3,333 records. Performance on this dataset may not exactly represent performance on new customers, future customer populations, or another telecom company. External validation would increase confidence in the model. **4. Limited feature information:** The model is restricted to the variables available in the dataset. Additional information such as customer complaints, payment history, satisfaction scores, contract type, previous retention offers, and changes in usage could potentially improve prediction. **Improvement Plan:** The analysis can be improved by collecting additional data, testing class-weighted models, evaluating SMOTE or other resampling methods inside leakage-safe cross-validation pipelines, optimizing the classification threshold, engineering additional features, evaluating XGBoost or other advanced boosting methods, and using feature-importance and explainability techniques.

1.6 Model Limitations and Next Steps

The analysis demonstrates that ensemble tree-based models are effective for predicting telecom customer churn, but the selected Random Forest model is not perfect. Its approximately **92.8% accuracy** and **0.724 F1-score** demonstrate strong overall performance, but its approximately **64.95% recall** indicates that a substantial number of actual churners are still missed. The main limitation is therefore the model's ability to identify all potential churners. Since the business objective is to identify customers who may leave, false negatives are particularly important. A customer who churns but is predicted as a non-churner cannot be targeted by a retention strategy. Another limitation is the class imbalance in the dataset. Only 483 of 3,333 customers belong to the churn class. Therefore, additional evaluation techniques and imbalance-handling strategies should be considered before deploying the model in a real business environment. **Recommended Next Steps**

1. Optimize the classification threshold to investigate whether recall can be improved without an unacceptable reduction in precision.
2. Experiment with class-weighted models to give greater importance to churn customers.
3. Test SMOTE or other resampling techniques using a leakage-safe pipeline.
4. Evaluate additional algorithms, particularly XGBoost and other gradient-boosting approaches.

5. Perform additional feature engineering using customer behavior and usage information.
6. Investigate feature importance and model explainability to understand why customers are predicted to churn.
7. Analyze false positives and false negatives to understand where the model makes mistakes.
8. Validate the model on new external data to determine whether it generalizes beyond the current dataset.
9. Monitor model performance over time if it is eventually deployed. Overall, the current analysis provides a strong starting point, with **Random Forest being the preferred model based on the current test results and cross-validation performance**. Further work is required before considering the model production-ready, particularly around class imbalance, recall improvement, threshold optimization, and validation on additional data.